

Validity limits of time-averaged likelihoods for macroscopic kinetic inference

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Abstract Time-averaged measurements pervade biology, and the likelihoods used to interpret them are approximations whose reliability is rarely tested. We ask where such an approximation can be trusted and how it fails when it cannot. For macroscopic ion-channel currents, we simulate a two-state channel under a concentration jump and use the exact simulation as ground truth to judge a family of Gaussian likelihood approximations (non-recursive, recursive, interval-averaged, and their combinations). We measure faithfulness with likelihood-level diagnostics: standardized residuals, the score, and an information-distortion matrix that compares the covariance of the score with the model's own Fisher information. We then map how each approximation deforms inference as a function of the number of channels, the acquisition interval, and the instrumental noise. Only the interval-averaged recursive likelihood (MacroIR) stays calibrated across the practical regime; the others lose or misallocate information in ways that trace to the two Gaussian approximations they make, one on channel occupancy and one on the single-interval signal. We locate the multinomial, telegraphic, and Gaussian regimes where the approximations degrade, and we show the distortion carries a concrete consequence: it maps to a correction of the Bayesian evidence that ranks competing mechanisms. The result is a validity map for time-averaged likelihoods and a calibrated default, MacroIR, for macroscopic recordings.

Introduction

The macroscopic interval likelihood and its approximations

A macroscopic current is the summed signal of a population of ion channels, each of which behaves as an independent continuous-time Markov chain over a small set of conformational states. We write N_{ch} for the number of channels and K for the number of kinetic states of one channel ($K = 2$ throughout this paper, carried symbolically where it costs nothing). Each channel is governed by a generator (rate) matrix $\mathbf{Q} \in \mathbb{R}^{K \times K}$, whose transition matrix over an elapsed time t is the row-stochastic $\mathbf{P}(t) = e^{\mathbf{Q}t}$ with entries $P_{i \rightarrow j}(t)$. A channel in state k carries a conductance γ_k , and we collect these in the vector $\boldsymbol{\gamma} \in \mathbb{R}^K$.

The likelihood we need is the probability of a recorded sample given the kinetic parameters and the data seen so far. Its exact form would require the distribution of the interval-averaged conductance of the whole population, which has no closed expression. Every practical likelihood replaces that distribution with a Gaussian, and the members of the family studied here differ only in what the Gaussian is conditioned on. The rest of this section unpacks that sentence.

The observable is an interval average

A recorded sample is the current averaged over the acquisition window rather than sampled at an instant. The acquisition electronics integrate the current across the sampling window Δ and pass it through an anti-aliasing filter before it is written down, so the quantity a likelihood must describe is the windowed average

$$\bar{y}_t = \frac{1}{\Delta} \int_{t-\Delta}^t y(\tau) d\tau + \eta_t, \quad (1)$$

where $y(\tau)$ is the instantaneous population current and η_t is the instrumental noise accumulated over the window. This interval average is the physical observable. When Δ is much shorter than the fastest relaxation of $\mathbf{Q}$ the average sits close to the instantaneous current and the distinction is negligible; when Δ is comparable to or longer than that relaxation the two diverge, and a likelihood built on instantaneous sampling describes the wrong random variable.

Two Gaussian approximations

Each member of the family makes two closures, on two different objects.

The two Gaussian approximations

Occupancy closure. The joint distribution of the channel occupancies is exactly multinomial when the channels are independent, which is also the maximum-entropy distribution consistent with the mean occupancies. It is replaced by a multivariate Gaussian with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, valid for large N_{ch} by the central limit theorem and degrading as the channel count falls.

Interval-signal closure. The distribution of the interval-averaged conductance over one window is a telegraph-like object with no closed form. It is replaced by a Gaussian, valid when the window spans many transitions or when instrumental noise dominates, and degrading as the window shrinks below the relaxation time.

The higher moments these closures discard do not vanish. They reappear in the data as temporal correlation that the likelihood does not predict, which is what the correlation term of the information-distortion decomposition measures (Diagnostics). Three regimes follow, and we name them here so the maps in the Results have vocabulary: the *multinomial* regime (few channels, where the occupancy closure fails), the *telegraphic* regime (windows far shorter than the relaxation time, where the interval-signal closure fails), and the *Gaussian* regime, where both closures hold because the channel count is high enough for the occupancy Gaussian while the window is long enough, or the instrumental noise large enough, for the interval-signal Gaussian. Any Gaussian likelihood of this kind must fail somewhere. The aim of this paper is to show that the failures are the predicted degradation of these two closures rather than defects of a particular algorithm.

The family as a coordinate system

The family has two knobs. One axis is the window: how much of the acquisition interval the conductance is conditioned on. The other is recursion: whether the occupancy distribution is conditioned on the data as the recording proceeds.

Along the recursion axis, a recursive (prequential) member factorizes the likelihood as $\prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$, so the occupancy distribution carried into each window is the posterior left by the previous windows and the hidden state is conditioned on the data. A non-recursive (open-loop) member factorizes as $\prod_t m_t(\bar{y}_t)$, with each window's predictive propagated from the initial condition and never conditioned on the observed trace.

Along the window axis the members form a ladder, indexed by how many of the interval's endpoints the conductance is conditioned on (Table 1).

Conditioned on	Members	Conductance model
no endpoints	NR, R	instantaneous; the acquisition averaging is ignored
one endpoint (the start)	NMR, MR	interval-mean conductance given the start state
two endpoints (the boundary)	IR	interval-mean conductance given both boundary states; inter
the full trajectory	(exact)	intractable; supplied by the stochastic simulation as ground t

Table 1. The window axis as an endpoint ladder. Crossing it with the recursion axis gives the implemented grid: NR (non-recursive, av = 0), R (recursive, av = 0), NMR (non-recursive, av = 1), MR (recursive, av = 1), IR (recursive, av = 2), where av counts the conditioned endpoints. MacroIR is IR.

Two things follow from the ladder. First, MacroIR occupies the top rung below the intractable exact likelihood, so its standing in the family is structural and set before any measurement. Second, the lower rungs are established likelihoods rather than constructed foils: the non-recursive instantaneous member is the independent-interval likelihood of *Milescu et al. (2005)*, and the recursive instantaneous member is the macroscopic recursive filter of *Moffatt (2007)* and the generalized filter of *Münch et al. (2022)*. The family is one coordinate system that contains these methods as special cases.

The boundary state

A *boundary state* is the pair (i_0, i_t) of a channel's states at the two ends of an acquisition interval. Chaining intervals, the end state of one interval is the start state of the next, so there is a single state variable per junction: the filter conditions on the states at the interval boundaries and marginalizes the trajectory between them analytically. This is the same device as static condensation, or the spatial Markov property, applied on the time axis. It is not a transition state in the mechanistic sense, and we keep the word *transition* away from it. Because the joint dynamics of the two endpoints are generated by the Kronecker sum $\mathbf{Q} \oplus \mathbf{Q} = \mathbf{Q} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{Q}$, whose propagator factorizes as $e^{(\mathbf{Q} \oplus \mathbf{Q})t} = e^{\mathbf{Q}t} \otimes e^{\mathbf{Q}t}$, the moments over the K^2 boundary pairs collapse to bilinear forms in the K -dimensional state space and no explicit K^2 object is formed (Methods).

The ladder needs two conductance objects, both built from the boundary-conditioned single-channel moments $\bar{\Gamma}_{i_0 \rightarrow i_t}$ (the mean interval-averaged conductance of a channel that starts in i_0 and ends in i_t) and $\bar{V}_{i_0 \rightarrow i_t}$ (its variance), which are computed from $\mathbf{Q}$ and the recording filter (Methods). The start-conditioned mean conductance is the K -vector

$$(\bar{\gamma}_0)_{i_0} = \sum_{i_t} P_{i_0 \rightarrow i_t}(\Delta) \bar{\Gamma}_{i_0 \rightarrow i_t}, \quad (2)$$

which conditions on the state at the interval's start alone and is the object used at av = 1. The boundary-conditioned mean conductance is the same average retained on both endpoints, so it stays a $K \times K$ object rather than collapsing to a vector, and is the object used at av = 2.

The interval update

Over a window of length Δ , the occupancy mean and covariance propagate as

$$\boldsymbol{\mu}^{\text{prop}} = \mathbf{P}(\Delta)^T \boldsymbol{\mu}_0, \quad (3)$$

$$\boldsymbol{\Sigma}^{\text{prop}} = \mathbf{P}(\Delta)^T (\boldsymbol{\Sigma}_0 - \text{diag}(\boldsymbol{\mu}_0)) \mathbf{P}(\Delta) + \text{diag}(\boldsymbol{\mu}^{\text{prop}}), \quad (4)$$

where $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ is the occupancy Gaussian at the start of the window. The predicted mean of the interval-averaged current is

$$\bar{\mathbf{y}}^{\text{pred}} = N_{\text{ch}} \boldsymbol{\mu}_0^T \bar{\boldsymbol{\gamma}}_0, \quad (5)$$

and its predicted variance decomposes into three independent contributions,

$$\sigma_{\text{pred}}^2 = \epsilon_{\Delta}^2 + N_{\text{ch}} \boldsymbol{\gamma}^T \widetilde{\boldsymbol{\Sigma}} \boldsymbol{\gamma} + N_{\text{ch}} \boldsymbol{\mu}_0^T \boldsymbol{\sigma}_{\bar{\gamma}_0}^2. \quad (6)$$

The first term ϵ_{Δ}^2 is the instrumental noise over the window, which carries the acquisition averaging and decreases with Δ ; its parametric form is given in Methods. The second term is the population gating variance, the fluctuation of the window current transmitted through the occupancy covariance: $\gamma^T \Sigma \gamma$ is the boundary covariance of the window collapsed onto the conductance, computed without ever forming the K^2 boundary matrix. The third term is the within-window per-channel variance, with $\sigma_{\gamma_0}^2$ the start-conditioned interval variance built from $\bar{V}_{i_0 \rightarrow i_t}$.

Writing $\mathbf{g} = \text{Cov}(\text{end-of-window occupancy}, \bar{y}) / N_{\text{ch}}$ for the per-channel cross-covariance between the interval current and the occupancy at the end of the window, and $\delta = \bar{y}^{\text{obs}} - \bar{y}^{\text{pred}}$ for the innovation, the Gaussian update of the occupancy is the scalar Kalman correction

$$\mu^{\text{post}} = \mu^{\text{prop}} + \frac{\mathbf{g}}{\sigma_{\text{pred}}^2} \delta, \quad (7)$$

$$\Sigma^{\text{post}} = \Sigma^{\text{prop}} - \frac{N_{\text{ch}} \mathbf{g} \mathbf{g}^T}{\sigma_{\text{pred}}^2}. \quad (8)$$

The factor N_{ch} enters the covariance downdate but not the mean correction because Σ is carried on the population scale, the same N_{ch} that multiplies the gating variance in Eq. 6, whereas μ and the gain $\mathbf{g}$ are per-channel objects, so the rank-one correction to Σ must be rescaled by N_{ch} to sit on its scale while the correction to μ needs no such factor. In a recursive member the posterior pair $(\mu^{\text{post}}, \Sigma^{\text{post}})$ becomes the prior for the next window; in a non-recursive member it is not carried forward, and each window's predictive is propagated from the initial condition alone. Reusing the posterior as the next prior is itself a closure: it is exact only while the end marginal is a sufficient summary of the boundary posterior, which the Gaussian projection does not guarantee. For this reason the recursive members' Gaussian information is a model-based quantity and is not claimed to be the exact information of the process; how far it departs from the truth is what the Diagnostics measure.

The single algebraic difference within the family sits in the conductance object of Eq. 6. MacroIR conditions the interval-averaged conductance on both boundary states; MacroMR conditions it on the start state alone, using $\bar{y}_0$ of Eq. 2 in its place. This drops the boundary cross-covariance term

$$N_{\text{ch}} \gamma^T \Sigma \gamma \quad (9)$$

that MacroIR retains, and it is the only place the two members diverge. Its consequence for the predicted variance across the regime grid is characterized in the Results.

The top of the ladder

The exact likelihood conditions on the full trajectory inside each window and is intractable, which is why the two closures above are needed at all. The same stochastic model that makes the exact likelihood intractable can be simulated forward exactly, one trajectory at a time, so the simulation realizes the object at the top of the ladder and serves as the ground truth against which every rung below it is judged.

Diagnostics: testing a likelihood against its simulation

An approximate likelihood is a misspecified model, and misspecification leaves a classical signature. At the true parameters the score no longer has zero mean, so the estimator the likelihood defines is biased; and the covariance of the score no longer equals the negative expected curvature, so the uncertainty the likelihood reports is wrong. These are the two Bartlett identities, and their failure is the content of the information-matrix-equality diagnostic of [White \(1982\)](#) and of the sandwich covariance of [Huber \(1967\)](#); [Godambe \(1960\)](#). In the classical setting those identities can only be tested, through a statistic with a known finite-sample defect, because the data-generating law is unknown. Our setting removes that limitation. The forward process is an exact continuous-time Markov chain (CTMC) and we simulate it exactly, so we hold unlimited data drawn from the very law the likelihood claims to describe, at parameters we set. The classical test therefore becomes

a direct measurement: the anchor curvature is computed analytically and the score covariance is estimated by Monte Carlo over independent simulated recordings, with no null hypothesis to reject. The apparatus is standard; what is new here is applying it in this domain and reading its output as a measured property of each approximation rather than as a hypothesis to accept or reject.

We read three quantities off the simulation, in increasing strength. The predictive residual and the information anchor are evaluated at the estimate; the expectation behind the score bias is taken at the simulation truth. For a misspecified likelihood the optimum and the truth differ, and that optimum-versus-truth gap is what makes the bias observable.

The first is the standardized residual. With μ_t and σ_t^2 the likelihood's own predictive mean and variance for interval t , the residual $r_t = (y_t - \mu_t)/\sigma_t$ must have mean zero, unit variance, and no autocorrelation across intervals if the one-step predictive distribution is correct (Figure 3). This probes the predictive one interval at a time. It is the cheapest of the three and the only one an experimentalist can run on a real recording, since it needs the predictive moments but not the true parameters. Residual whiteness in particular is not optional: residual autocorrelation is the same temporal signal that the strongest check quantifies below.

The second is the score. Writing $\ell(\theta)$ for the total log-likelihood and $S(\theta) = \nabla_\theta \ell(\theta)$ for its gradient, correct specification requires $\mathbb{E}[S(\theta^*)] = 0$ at the true parameters θ^* . A nonzero mean score is what biases the estimate, and projected through the parameter covariance it becomes a displacement in parameter units, $b = \mathbf{H}^{-1} \mathbb{E}[S]$, the reader-facing statement of how far the estimate is pushed (Figure 3). This projection is a first-order approximation, from the expansion of the score at the optimum given below. Unlike the residual, the check needs a known truth, so it requires the simulator, and its confidence interval narrows as the number of simulated recordings grows.

The third and strongest is the information itself. Correct specification requires $\text{Cov}[S] = \mathbf{H}$, the information the likelihood reports. An approximation can pass the residual and score checks and still violate this one by an order of magnitude, and NMR does exactly that: it averages the conductance across the interval, so its per-interval predictive is unbiased, yet the uncertainty it reports is far smaller than the estimate actually delivers. NR fails on both axes: it uses the instantaneous conductance, so its per-interval moments are wrong to begin with; that biases the score, and it is over-confident on top of that. Averaging governs whether the estimate is biased, recursion governs whether the reported information is inflated, and holding those two failures apart is the reason this section exists. Because the failure is directional, we record it as a matrix,

$$\mathbf{C} = \mathbf{H}^{-1/2} \mathbf{J} \mathbf{H}^{-1/2},$$

with $\mathbf{J} = \text{Cov}(S)$ the covariance of the total score, cross-interval terms included. This definition is exact algebra given $\mathbf{H}$ and $\mathbf{J}$, and under correct specification $\mathbf{J} = \mathbf{H}$ so $\mathbf{C} = \mathbf{I}$ exactly. We fix the direction by convention: $C_{ii} > 1$ means the likelihood under-reports the uncertainty in parameter i and is over-confident, $C_{ii} < 1$ means it over-reports. The eigen-directions of $\mathbf{C}$ say which combinations of parameters are misreported, which a scalar summary cannot recover.

The anchor $\mathbf{H}$ is the likelihood's own Gaussian Fisher information, $\mathbf{H} = \sum_t \mathbb{E}[(\nabla \mu_t)(\nabla \mu_t)^\top / \sigma_t^2 + (\nabla \sigma_t^2)(\nabla \sigma_t^2)^\top / \sigma_t^4]$ formed from the same predictive moments and their parameter derivatives that the score already requires, at no extra cost. For a macroscopic algorithm the likelihood is Gaussian by construction, so this is the model's own Fisher and $\mathbf{C}$ is the information-matrix-equality test carried out in the model's own frame. Two properties follow. The Gaussian Fisher is positive semidefinite by construction, so the diagnostic never has to be rescued from a numerically indefinite anchor. And it is analytic, so it does not inherit the finite-difference noise of a numerical Hessian, which in this regime is severe: the finite-difference Fisher is widely indefinite per replicate, and only pooling across replicates recovers a definite matrix. By design the anchor is the Gaussian Fisher; the numerical Fisher we compute alongside it serves to gauge how faithful that Gaussian anchor is. Comparing the approximation to its own Fisher is not circular. $\mathbf{H}$ is the information the likelihood

claims, $\mathbf{J}$ is the information it delivers against data from the true process, the two are computed from different objects, and their disagreement is the definition of misspecification.

The same matrix repairs what it measures. The sandwich covariance

$$\Sigma = \mathbf{H}^{-1} \mathbf{J} \mathbf{H}^{-1} = \mathbf{H}^{-1/2} \mathbf{C} \mathbf{H}^{-1/2}$$

is the corrected parameter covariance: it substitutes the information the likelihood actually carries for the information it reported. This step, and the bias vector b above, are first-order approximations, from a Taylor expansion of the score at the optimum with the empirical Hessian replaced by its expectation $-\mathbf{H}$ (see Supplementary Information). We verify the correction directly, without leaning on the expansion, by comparing the sandwich-corrected ellipse against the empirical covariance of the maximum-likelihood estimates (MLEs) over the simulated recordings (Figure 2). A diagnostic that also corrects the error bars it flags is a stronger result than one that only scores algorithms.

The distortion splits into two physically distinct parts (Figure 6). The sample distortion $\mathbf{C}_s = \mathbf{H}^{-1/2} \mathbf{J}_s \mathbf{H}^{-1/2}$, built from the within-interval score covariance $\mathbf{J}_s = \sum_t \text{Cov}(s_t)$ with $s_t = \nabla_{\theta} \ell_t$ the score of interval t , measures per-sample departure from Gaussianity; to the leading corrections it is the third and fourth cumulants of the interval current, the non-Gaussian shape the Gaussian predictive cannot carry. The correlation distortion $\mathbf{R} = \mathbf{J}_s^{-1/2} \mathbf{J} \mathbf{J}_s^{-1/2}$ measures what the within-interval covariance misses, the cross-interval correlation of the score, which is exactly the local time correlation of the current that *Milescu et al. (2005)* named as the dominant error of non-recursive fitting; $\mathbf{R} = \mathbf{I}$ when the score carries no cross-interval dependence. These compose exactly. The exact identity is

$$\mathbf{C} = \mathbf{K} \mathbf{R} \mathbf{K}^T, \quad \mathbf{K} = \mathbf{H}^{-1/2} \mathbf{J}_s^{1/2},$$

which holds as algebra for any $\mathbf{H}$ and $\mathbf{J}_s$, since substituting the definitions collapses the chain. The factor $\mathbf{K}$ satisfies $\mathbf{K} \mathbf{K}^T = \mathbf{C}_s$, so it is a legitimate square-root factor of the sample distortion, but it is not the symmetric principal square root, and the form $\mathbf{C} = \mathbf{C}_s^{1/2} \mathbf{R} \mathbf{C}_s^{1/2}$ holds only where $\mathbf{H}$ and $\mathbf{J}_s$ commute. What composes with no assumption at all is the information volume,

$$\log \det \mathbf{C} = \log \det \mathbf{C}_s + \log \det \mathbf{R},$$

an exact identity because the determinant is multiplicative. The per-parameter diagonal, the trace, the eigenvalues and the Frobenius norm do not compose across the split; only the volume does, and only the volume is decomposed in the maps.

The correlation distortion has a reader-facing translation. For any additive statistic X_t with sum $L = \sum_t X_t$, the ratio $\kappa = \text{Var}(L) / \sum_t \text{Var}(X_t)$ is unity when the terms are independent and larger when they are positively correlated, and it converts to an effective sample size $T_{\text{eff}} = T/\kappa$: a non-recursive likelihood believes it has T independent intervals when it effectively has T/κ . This is the number from the diagnostic most worth quoting, and it is exact by definition. When $\mathbf{R} \approx \kappa \mathbf{I}$ the temporal error is isotropic and this scalar captures it; when $\mathbf{R}$ is anisotropic, or when $\mathbf{C}_s \neq \mathbf{I}$, the distortion cannot be reduced to a sample-size correction and the full matrix is needed.

Two conventions govern the edge cases. The direction is fixed as above, $C_{ii} > 1$ being over-confidence, stated once so every verdict reads the same way. And the anchor can be near-singular where a parameter is unidentifiable, which by design happens once the open population relaxes (Figure 4); there we work in the retained eigenspace of $\mathbf{H}$ above a numerical cutoff and leave the quantity undefined, not finite, when that eigenspace is empty. Undefined is not zero, and the maps render those cells in grey to keep the two apart.

Results

The results follow one order, in which each figure answers the objection the previous one raises. The five approximations differ visibly in a single filter step (Figure 1); that difference is miscalibrated parameter uncertainty (Figure 2); the miscalibration lives in the likelihood itself, interval by interval

(Figure 3); and it comes from correlation across time rather than any per-sample error (Figure 4). The last two subsections then map how large the distortion is across the operating regime and decompose why it is large where it is.

A minimal two-state model and the algorithm family

We keep the model at the smallest size that still poses the estimation problem, so that any miscalibration we measure is a property of the likelihood approximation and not of the model. A patch of N_{ch} identical channels, each switching between a closed and an open state (closed $\rightleftharpoons$ open) at an agonist-dependent opening rate k_{on} and a closing rate k_{off} , is driven by a step change in agonist concentration; only the open state conducts. The recorded current is the patch current averaged over each measurement interval, with white instrumental noise added. That interval average is the object every approximation in this paper has to account for. Exact stochastic simulation of the channel population supplies the ground truth, because the forward problem of generating a trajectory is tractable where the inverse problem of writing down the exact likelihood of an averaged recording is not.

The five approximations are one recursive filter run under two settings, and they form a ladder in how much of each acquisition interval the hidden state is conditioned on. The recursion setting fixes whether the occupancy covariance is carried from one interval into the next as the prior for the next (R, MR, IR) or each interval is scored afresh from the same fixed start (NR, NMR). The averaging setting fixes the conductance model: an instantaneous conductance (NR, R), the interval-mean conductance given the state at the interval's start (NMR, MR), or the interval-mean conductance given the channel's states at both ends of the interval, which we call boundary-conditioned (IR). MacroIR is the boundary-conditioned recursive member, the top rung below the intractable exact likelihood. Figure 1 lays one turn of the filter cycle side by side for all five, the prior, the predicted current, the Bayesian update and the accumulated log-likelihood; it makes the coordinate system concrete and carries no claim about accuracy.

Which approximations are calibrated: parameter recovery

Figure 2 turns the family into a statement about parameter uncertainty. From ten thousand independent recordings of a patch of $N_{\text{ch}} = 100$ channels, each of the four free parameters (the rates k_{on} and k_{off} , the unitary current i , and the channel number N_{ch}) is re-estimated by maximum likelihood under each approximation, and the spread of those maximum-likelihood estimates is compared with the uncertainty each approximation reports from its own Fisher information. The two are different objects. For the non-recursive members the Fisher information is far too tight, understating the true spread by roughly a factor of 10 to 16. The recursive members are much closer: R sits within about 1.2 to 1.4, and MR, which conditions on only one end of the interval, is still over-confident at about 1.5 to 2.1. IR alone is calibrated: its reported and realized uncertainties coincide across every one of the four parameters and it carries no detectable bias, so the ellipse it draws is the ellipse the estimates occupy.

The ordering follows the endpoint ladder. The members that ignore the acquisition window over-state what the data determine; the member that conditions on both ends of the interval does not, and the same over-confidence is not shared once the interval is respected. The distortion is measurable and also removable. Rescaling each reported covariance by the observed covariance of the score, the sandwich correction, brings every approximation, IR included, back to a spread ratio near one. That one correction repairs all five is what makes the distortion a usable quantity rather than a complaint.

A natural worry is that the clouds are driven by a handful of atypical recordings. They are not. Within each cloud the maximized log-likelihood of a recording is essentially uncorrelated with how far its estimate lands from the truth (Figure S2), because the log-likelihood measures how atypical a realization's noise and gating happened to be, not how well its parameters were recovered.

These recovery panels (Figures 2, S2 and S3) are computed on the numerical, finite-difference Fisher information of the 433ed13 simulation run; the definitive versions on the analytic Gaussian-Fisher anchor are in preparation. Per interval, the variance of the score matches the per-step Fisher information for all five members (Figure 4).

Time-resolved calibration and the score/Fisher checks

The miscalibration is present in the likelihood itself, interval by interval, and is not an artifact of the fit. Scoring one common set of a thousand recordings interval by interval (Figure 3) shows first that the recursive members reach a total log-likelihood near -145 while the non-recursive pair reaches about -230 , a gap of roughly 85 nats: under the recursive, conditional likelihoods the same data are far more probable.

The rest of the figure is the hinge of the argument. Per interval, every one of the five reports its information correctly: the variance of the per-interval score matches the per-interval Fisher information, $J_i/F_i \approx 1$, for all of them. The failure appears only once the information is accumulated. The ratio of the variance of the summed score to the summed information stays near one for the recursive members and diverges for the non-recursive pair. So the local information content is right everywhere; the error is entirely in how the information adds up over time.

The reason is visible in the same figure. The per-interval score of the non-recursive members is correlated across time, and a correlated score makes the variance of its sum exceed the sum of the per-interval informations. This temporal correlation of the current, which the recursive filters remove by conditioning on the past, is the empirical form of the diagnosis that Milescu and colleagues gave for the same failure *Milescu et al. (2005)*.

Where the information lives: the Fisher falls to zero on relaxation

Two results follow from resolving the per-step information in time (Figure 4). The first closes the mechanism. The per-step identity between score variance and Fisher information holds for all five approximations, the naive members included: at each interval the two agree within the bootstrap interval. The over-confidence of the non-recursive members is therefore entirely the cross-time correlation of the score. Every approximation gets each interval right; they differ only in how the intervals combine.

The second result is a property of the measurement, not of any one filter. For the recursive filters the information about the channel number N_{ch} , the opening rate k_{on} and the unitary current i drops to zero the moment the agonist is removed and the open population relaxes: having conditioned on the past, the filter gains nothing further from watching the channels decay. The closing rate k_{off} is the exception, staying informative through the relaxation, which its own kinetics govern. Because this behaviour belongs to the macroscopic observable and not to the approximation, it survives every member of the family. Reading off the likelihood when each parameter stops being measured is exactly the question that a recent treatment of this class of models leaves open *Del Core and Mirams (2025)*.

A validity map across channels, interval, and noise

The recovery and time-resolved panels fix one operating point; the deliverable is the whole map. Sweeping the channel number, the acquisition interval and the instrumental noise, and asking in each cell whether the likelihood is self-consistent, its reported uncertainty within tolerance of the realized spread, sorts the family cleanly. IR is self-consistent over almost the entire plane and improves as noise rises; R holds over a large but incomplete region that also grows with noise; MR holds only in a small corner; and the non-recursive pair is self-consistent essentially nowhere.

[TODO] Fig. five pending Gaussian fill: self-consistency fractions and final panel

Where IR does depart from calibration, at the fewest channels, it departs in the conservative direction: it under-states rather than over-states what the data determine, which is the harmless

direction for a user to be wrong in. The window-ignoring approximations fail the other way, inflating the information the data carry. This is the one asymmetry a practitioner needs, and it runs in IR's favour.

The map is a gradient, not a set of bordered territories. The multinomial, telegraphic and Gaussian regimes named in the theory are the asymptotic corners of the two Gaussian approximations; they set the direction of the gradient rather than drawing lines on the plane, and a cell is more or less distorted, never simply inside or outside a regime.

One reading of the map gives an experimenter something to change. Concentrating the same channels into fewer, larger patches costs almost nothing for the closing rate, more for the opening rate, and a great deal for the unitary current, so how channels are apportioned across patches trades directly against which parameters can be recovered. **[TODO]** Fig. five pending Gaussian fill: channel-pooling loss factors

Decomposing the distortion: correlation versus sample

The shape of the map has a mechanism. The information distortion separates into two parts, one from the correlation of the signal across intervals and one from the non-Gaussian shape of a single interval's averaged signal, the sample term, and the two failure modes live in different corners. The separation is exact only for the information volume: the log-determinant of the distortion is the sum of a correlation log-determinant and a sample log-determinant, with no residual. The full distortion matrix combines the two through a two-sided congruence, so the volume splits exactly while the trace and eigenvalue pieces carry a residual. The non-recursive members blow up in the correlation term at short intervals, where successive samples are strongly dependent, and in the sample term at coarse intervals, where a single interval's averaged signal is least Gaussian. **[TODO]** Fig. six pending Gaussian fill: decomposition grid Two Gaussian approximations, each failing where the theory says it should.

The sample term behaves in a way worth stating on its own. It is non-monotonic in the channel number: the single-interval non-Gaussianity falls as channels are added and the central-limit theorem takes hold, while the gating fraction of the signal variance rises, and the product of the two opposing trends leaves a worst channel count in between. Instrumental noise adds a pure-variance pedestal that dilutes the single-interval non-Gaussianity, so more noise makes the per-sample Gaussian approximation better. **[TODO]** Fig. six pending Gaussian fill: sample-peak location and its noise scaling We report this as a measurement; we do not present the fitted scaling as a confirmed law.

One feature of the map is easy to misread, and it belongs in the text rather than a caption. The large correlation distortion at the shortest intervals reflects the correct cost of oversampling: when the acquisition interval is far shorter than the channel's correlation time, successive samples carry redundant information, and a faithful likelihood has to register that redundancy. The left edge of the map shows the theory working as intended.

Discussion

Materials and methods

Minimal model and simulation

We work with a two-state channel scheme, closed (C) and open (O), the smallest scheme in which the acquisition-averaging effect we study is present. The closed-to-open transition is agonist dependent with association constant k_{on} multiplying the agonist concentration; the open-to-closed transition is agonist independent with dissociation rate k_{off} . Only the open state conducts, and the code applies a sign flip so the single-channel current is inward. At the start of every recording all channels are closed, so the occupancy is $(1, 0)$.

The scheme carries six parameters, all estimated in base-ten logarithmic coordinates. The values used to generate the data reported here, read from the production run scripts rather than

from the parameter tables in the repository (the tables hold stale values the figures never used), are $k_{\text{on}} = 10 \mu\text{M}^{-1} \text{s}^{-1}$, $k_{\text{off}} = 100 \text{s}^{-1}$, a unitary conductance g set to 1, and a current baseline b set to 1 (both in units of g), while the white-noise level and the mean channel number were swept (below). Current is expressed throughout in units of the unitary conductance g , which corresponds physically to a single-channel current of roughly 1 pA. The dissociation time $\tau = 1/k_{\text{off}} = 10 \text{ms}$ sets the natural time scale that normalises the acquisition-interval axis. The relaxation at the agonist condition is faster, with rate $k_{\text{on}}[A] + k_{\text{off}} = 200 \text{s}^{-1}$; the axis uses $\tau = 1/k_{\text{off}}$ by convention.

The emission model is white instrumental noise only. Both the pink (one-over- f) and the proportional noise channels are set to zero, so the term the reader may expect from flicker noise is genuinely absent. For a measurement interval t of duration Δ_t , the predicted current has mean $\bar{y}_t = N_{\text{ch}} \mu_t \gamma_t + b$ and variance

$$\sigma_t^2 = e_t + N_{\text{ch}} v_t, \quad e_t = \text{Current_Noise} \frac{f_s}{n_{\text{samp},t}} = \frac{\text{Current_Noise}}{\Delta_t},$$

where N_{ch} is the channel number, μ_t the occupancy mean, γ_t the interval-averaged conductance, f_s the raw sampling rate, $n_{\text{samp},t}$ the number of raw samples averaged into the interval, e_t the white instrumental variance over the interval, and $N_{\text{ch}} v_t$ the channel-gating variance. The white variance e_t falls as the interval lengthens, because more raw samples are averaged into each recorded value. The gating variance v_t is the single-interval conductance variance, whose form differs across the algorithm family and is derived in the Theory section.

The stimulus is a single concentration jump in three segments, a pre-pulse at 0 μM , an agonist step at 10 μM , and a post-pulse at 0 μM , with raw sampling at $f_s = 50 \text{kHz}$ throughout. Each recorded measurement is the mean of $n_{\text{samp},t}$ consecutive raw samples, so the averaging window is fixed at acquisition. That window is the acquisition interval the whole study is about, and its length is set by the number of raw samples entering each recorded value.

Ground truth is generated by exact simulation of the continuous-time Markov chain (CTMC). We use uniformization with 1,000 substeps per measurement interval, so the interval average of the conductance is realised from an actual sampled trajectory rather than from any moment-closure approximation. This is what licenses treating the simulation as the reference against which the Gaussian likelihood approximations are judged: the approximations are the object under test, and the simulator supplies the exact answer they are compared to. The two maximum-likelihood batteries used 10,000 recordings per grid cell; the time-resolved run used 1,000 recordings; the single-recording illustration (Figure 1) simulated one 20-channel trace with 250 within-interval substeps for visualisation and does not enter the quantitative comparison.

The simulation seed was set to 0, which in this engine is the sentinel for a random seed drawn from `std::random_device`, and the resolved seed was never logged. The deposited ensembles are therefore statistically equivalent to, but not bit-for-bit reproducible from, a rerun of the same script. Because each ensemble is large (1,000 to 10,000 recordings), the reported statistics are stable; the Monte-Carlo standard here is statistical equivalence, not bit identity.

The macroscopic likelihood approximations

The five likelihoods compared here are one filter run under two flags. The first flag, *recursive approximation*, sets whether the occupancy covariance is propagated between intervals. The second, *averaging approximation* $\text{av} \in \{0, 1, 2\}$, sets how the single-channel conductance is treated within an interval, and it literally counts the number of interval endpoints the hidden state is conditioned on.

Table 2 lists the members. Reading the endpoint flag as a ladder, the members condition the conductance on progressively more of the interval: no endpoints, one endpoint, then both boundary states, with the exact trajectory (what the simulator supplies) as the unreachable top rung. MacroIR is the top implemented rung: it conditions on the pair of channel states at the two ends of the interval, which we call the boundary state, and marginalises the interior analytically. It is the

Member	Recursive	av	Conditioning of the conductance
NR	no	0	instantaneous; the acquisition averaging is ignored
NMR	no	1	interval mean given the start state
R	yes	0	instantaneous; the acquisition averaging is ignored
MR	yes	1	interval mean given the start state
IR	yes	2	interval mean given both boundary states, interior marginalised

Table 2. The likelihood family as two flags. The members form an endpoint ladder: no endpoints (NR, R), one endpoint (NMR, MR), both boundary states (IR). The data key for NMR is `macro_NMR`.

likelihood introduced for the P2X2 receptor (*Moffatt and Pierdominici-Sottile, 2025*), here placed as one member of the completed grid, and it is verified in the Theory section to coincide with a time-integrated Kalman filter. The construction of the interval-averaged mean and variance, the boundary-state covariance, and the Kalman-like update are derived in the Theory section and its supplement; we do not restate them here.

Two reductions are worth stating because they anchor the family to prior work. Setting the boundary-conditioned conductance variance to zero in the recursive, no-endpoint member (R) recovers the macroscopic-fluctuation update of *Moffatt (2007)*, so R is the recursive ancestor already in print. The single structural difference between the M and I families is that the one-endpoint members drop the boundary cross-covariance term $N_{\text{ch}} \gamma^T \Sigma \gamma$ that IR retains.

Throughout the reported runs the remaining build flags were held fixed: *taylor variance correction* off, *micro approximation* off, and the *variance approximation* on. Because the Taylor variance correction is off, the recursive one-endpoint member that runs is MacroMR and not MacroMRT; the Taylor-corrected variants are not part of this paper.

The filter carries two numerical safeguards, present because the code runs them and because anyone reimplementing the filter meets the same constraint on day one. The occupancy mean is held on the probability simplex, and the occupancy covariance is kept positive semidefinite through a trust-region shrink of the Kalman-like mean update. Two constants enter, a binomial-count floor set to 5 and a minimum occupancy probability set to 10^{-12} .

Parameter estimation and the two anchors

Inference in each grid cell is two-stage. First, the ensemble is partitioned into groups and a separate maximum-likelihood estimate (MLE) is computed per group, producing a cloud of estimates whose scatter is the empirical parameter covariance. The optimiser is Gauss-Newton with Levenberg damping (initial damping 10, growth factor 3, maximum damping 10^{10} , at most 100 iterations, relative gradient tolerance 10^{-6} , value tolerance 10^{-10}), warm-started at the simulation truth. Two group sizes were used, 10 and 100; with 10,000 recordings these give 1,000 groups of 10 and 100 groups of 100. Second, a single joint MLE is computed over all replicates at once (one group of size n_{sim}), giving the pooled estimate θ_{pool} ; its near-zero gradient certifies that it is a stationary point of the pooled likelihood.

Warm-starting the per-group fits at the truth is deliberate. It isolates the error of the likelihood from the error of the optimiser, which is what this paper studies; it would be the wrong choice for a study of an estimator, which this is not.

The two anchors, the simulation truth θ_{sim} and the pooled fit θ_{pool} , are distinct points because the Gaussian macroscopic likelihood is misspecified with respect to the exact simulator. Their difference $\theta_{\text{pool}} - \theta_{\text{sim}}$ is the misspecification bias. Every diagnostic below is computed at both anchors, and the two answer different questions: at θ_{pool} the score is zero by construction so bias washes out, whereas at θ_{sim} the bias is visible but the Fisher information can be indefinite away from the maximum. Each figure states which anchor it uses.

Diagnostics and the Gaussian-Fisher anchor

Faithfulness of each approximate likelihood to the simulator is measured with likelihood-level diagnostics: standardized residuals, the score (the gradient of the log-likelihood), and an information-distortion matrix that compares the covariance J of the score across replicates with the model's own Fisher information. The definitions, sign conventions, and thresholds of these diagnostics are owned by the Diagnostics section and its metric registry; here we describe only how they were computed.

The Fisher information was formed two ways. The finite-difference construction, which we write F_{num} , is a central difference of the analytic score, with per-coordinate step $h_i = h_{\text{rel}} \max(|\theta_i|, 1)$ at $h_{\text{rel}} = 10^{-5}$, and it is symmetrised as $(F_{\text{num}} + F_{\text{num}}^T)/2$. Because the score itself is obtained by automatic differentiation, this is a single numerical derivative, not two. The analytic construction is the Gaussian Fisher information $\bar{G} = \sum_i \text{GFI}_i$, summed over intervals. This analytic Gaussian Fisher is the anchor the Diagnostics section writes $\mathbf{H}$, that is $\mathbf{H} = \bar{G} = \sum_i \text{GFI}_i$. Its per-interval term is

$$\text{GFI}_i = \frac{(\partial_{\theta} \bar{y}_i)(\partial_{\theta} \bar{y}_i)^T}{\sigma_i^2} + \frac{(\partial_{\theta} \sigma_i^2)(\partial_{\theta} \sigma_i^2)^T}{2 \sigma_i^4},$$

which follows from the analytic derivative alone and needs no extra likelihood passes. Comparing the score covariance J against either Fisher construction is the sandwich, or information-matrix-equality, diagnostic (**White, 1982**): equality between the Fisher and the score covariance ($\mathbf{H} = J$ under the analytic anchor, $F_{\text{num}} = J$ under the finite-difference one) holds only when the likelihood is correctly specified, and the extent to which they differ is what the distortion matrix quantifies.

Uncertainty on every reported statistic comes from a nonparametric bootstrap over groups, reported as percentile intervals at the 2.5, 16, 50, 84 and 97.5% quantiles. Despite the internal column names, no probit transform is applied; the intervals are ordinary empirical percentiles, verified equal to the R quantile of type 6. The diagnostic battery used 100 bootstrap replicates and a maximum lag of 10 for the score autocorrelation; the empirical-versus-theoretical distortion capstone used 200 replicates; and fewer than 10 groups leaves the interval slots empty (not-a-number filled), which is one source of empty cells in the maps. The raw per-group MLE cloud is written unbootstrapped; any resampling of it is done downstream in the figure code.

Two production batteries were run that differ only in which Fisher construction anchors the diagnostics. The finite-difference battery covers all five algorithms; the Gaussian battery covers MacroIR across the full channel-by-noise grid, with the remaining members being added on the canonical grid by the in-progress fill (below). Cross-algorithm comparisons are drawn from the finite-difference battery and MacroIR-specific mechanism results from the Gaussian battery. The split is a design choice rather than a completed cross-grid verification: the analytic Gaussian Fisher and the finite-difference Fisher target the same information, an identity the time-resolved figures show holds per interval, so anchoring a given panel on either construction is consistent. The finite-difference battery is retained to exhibit that agreement, not to anchor the mechanism results. The choice of anchor is recorded in the project decision log.

Regime grid and reduction

The acquisition interval was swept over seven levels of Δ/τ , namely 1, 0.5, 0.2, 0.1, 0.05, 0.02 and 0.01, realised as 500, 250, 100, 50, 25, 10 and 5 raw samples averaged per interval, that is step durations of 10, 5, 2, 1, 0.5, 0.2 and 0.1 ms, giving 10, 20, 50, 100, 200, 500 and 1,000 steps in the record. The total record duration is held fixed at 100 ms across the sweep, so the interval is varied without varying the amount of data. Within each record the pre-pulse, agonist and post-pulse segments hold a fixed ratio of 20:40:40 percent of the steps.

The swept “noise” axis is dimensionless. It is $\text{noise_in_conductance_tau} = 10 \text{Current_Noise } k_{\text{off}}/g^2$, the white current variance accumulated over one relaxation time τ taken relative to the single-channel signal power g^2 , with a factor placing the level at which the white noise equals the single-channel signal near a label of 10. With the run values it equals $1000 \text{Current_Noise}$, so the figure

label 0.1 corresponds to the engine parameter $\text{Current_Noise} = 10^{-4}$ (a white-noise power spectral density in units of $g^2 \text{ s}$). The figures carry the dimensionless label on their axes; the underlying engine parameter is not what is plotted.

The mean channel number N_{ch} and the noise level were the other two map axes. The finite-difference production grid on disk holds all five algorithms at $N_{\text{ch}} \in \{10, 100, 1000, 10000\}$ and noise labels $\{0.1, 1, 10\}$ at 10,000 recordings, with MacroIR additionally at $N_{\text{ch}} \in \{20, 50, 200, 500\}$; the Gaussian grid holds MacroIR at $N_{\text{ch}} \in \{10, 20, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$ and the six noise labels $\{0.05, 0.1, 0.2, 0.5, 1, 10\}$. Coverage of the four non-IR members on the Gaussian anchor is supplied by the in-progress D-0 fill, on the canonical $N_{\text{ch}} \in \{10, 100, 1000, 10000\}$ grid at the three noise labels $\{0.1, 1, 10\}$ (not MacroIR's six); until that fill lands these members are not yet deposited across the grid, and any figure that needs them refreshes when it does.

Each grid cell is reduced through six stages: the per-group MLE cloud; the pooled joint fit θ_{pool} ; the Fisher information at both anchors; the score at both anchors, whose across-replicate covariance is J ; the diagnostic battery paired with the Fisher at each anchor; and the empirical-versus-theoretical distortion capstone, anchored at θ_{pool} . Each cell writes five comma-separated-value families (the estimate cloud, the pooled fit, the two anchor batteries, and the capstone).

Reproducibility and computational environment

The engine is `macro_dr`, built as C++20, and each analysis is a script in a small typed domain-specific language run in a single process (Moffatt, 2025). Provenance is self-documenting at three levels: on every invocation the binary snapshots the assembled script and its full argument list into a run ledger; every analysis output file stamps the build's short git commit hash as its first row; and the data directories are named by that hash (433ed13 for the finite-difference battery, 1c2ae6f for the Gaussian battery), so results from different code versions can never overwrite one another and multi-commit provenance is documented per file.

The production batteries ran under SLURM at 32 CPUs per task and 48 GB of memory (96 GB for the parameter-recovery lane), with the linear-algebra back end pinned to a single thread so that OpenMP parallelises the per-simulation and bootstrap loops. The environment variable `MACRODR_AXIS_SERIAL` is load-bearing: it serialises the internal grid-combination loop, without which memory grows with the number of concurrent combinations and the jobs exhaust memory.

The one reproducibility limit worth stating twice is the simulation seed. Because the seed was the random sentinel and the resolved value was not recorded, the exact deposited ensembles cannot be regenerated bit for bit; a rerun yields a statistically equivalent, numerically different ensemble. All resampling that operates on a fixed dataset (the bootstraps, the group partitioning) is deterministic and reproducible given the data. Deposition of the data and code, with pinned versions, is stated in the Data and code availability section.

Data and code availability

Acknowledgements

Author contributions

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